# SEMINAR SUMMARY: MARCH 31, 2026 - NEAL BUSHAW BOOTSTRAP PERCOLATION ON GRIDS.

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Percolation theory is a branch of mathematics rooted in time based graphical shading changes related to a set of rules and an initial condition. We were presented with an $n \times m$ square tiling of the plane. Our initial condition was a collection of shaded blocks, take Figure 1 for example. The percolating mechanic was then set to being that if a square has $k$ shaded (or infected) neighbors then in the next time step, it will become shaded, and time steps in definite intervals. For example, if $k = 1$, the behaviour is as if each square becomes a "source" in the sense of water or sand, and over time the entire board is filled so long as there is a single shaded block at our initial condition.

We then began looking at the scenario for $k = 2$ (i.e. two neighbors, as edge neighbors, not corner neighbors). If we were to take Figure 1, and step it once in time, then we would get Figure

*Date*: April 2, 2026.

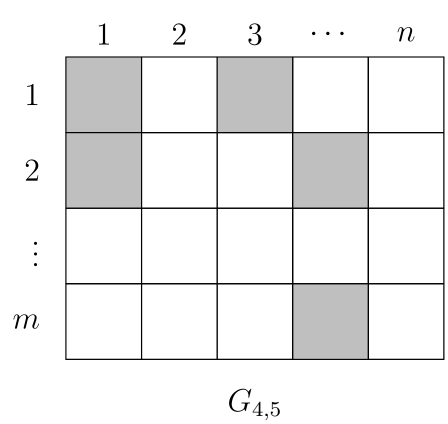


FIGURE 1. A rectangular $G_{4,5}$ tiled with squares, five shaded as bootstrap for percolation.

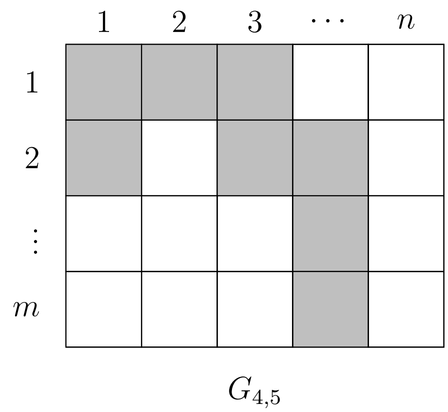


FIGURE 2. A rectangular $G_{4,5}$ tiled with squares, eight shaded after time step 1 with $k = 2$.